INVESTOR PRESENTATION · 2026

VELSANET

The Next-Generation AI-Native Network

Reimagining connectivity from the ground up — a fully integrated, intent-driven, optical-first network architecture for the AI era.

23	3	5	ITU
WHITE PAPERS	AI LAYERS	POLYHEDRAL TIERS	STANDARDS TRACK

THE PROBLEM

The Internet Is Structurally Broken

⚡ Traffic Explosion

IoT, streaming & AI workloads are outpacing legacy routing capacity.

🔒 Zero Structural Security

The Internet was designed for trusted environments — not adversarial ones.

🧩 Intent Gap

Networks carry bits, not meaning. Human intent & AI cognition are ignored.

🕸 Centralization Paradox

Every attempt to fix the Internet adds a new layer — compounding fragility.

The root cause: the Internet's packet-switching architecture was never designed for intelligence.

OUR VISION

A Living, Intelligent Network Ecosystem

Topology First

Space is defined before addresses — structure drives intelligence.

Optical-Native E2E

No packet switching. Physical optical cores for every connection.

Intent-Driven

Networks carry meaning, not just bits. AI cognition is embedded.

Sovereign Governance

Global coordination without centralized control — via projection.

CORE TECHNOLOGY

Breakthrough Innovations

01 Multi-Optical-Core Transceiver (MOCT)

Fixed photonic module, hundreds of parallel cores. No connectors, no ASIC routing — pure physical E2E.

02 Polyhedral Network Topology

T4→C6→O8→D12→I20: five geometric primitives form the grammar of network space.

03 Three-Layer AI (PAI / AAI / AsAI)

Personal, Agent & Assistant AI embedded natively into network nodes — not bolted on.

04 Cube Memory (T-C-I-E-M)

Five-axis semantic memory. AI retrieves meaning, not just data — and cubes transplant between nodes.

05 Spatial Genesis & Digital DNA

ARCHITECTURE

Hierarchical Intelligence by Design

- **Q2** Global Network AI (VELSA)
- **Q3** Continental Network AI
- **Q4** National Sovereign Layer
- **Q6** Regional Coordination
- **Q7** Execution Layer (Devices)

Higher layers project structural states — they do not command. Sovereignty is preserved at every tier.

AI ARCHITECTURE

Three-Layer Distributed Intelligence

AsAI Assistant AI — National & Global Scale

Oversees AAI cooperation across domains · prediction, policy integration · housed in Icosahedron (I20) nodes

AAI Agent AI — Regional Coordination

Coordinates PAI clusters · real-time orchestration · operates from Dodecahedron (D12) nodes

PAI Personal AI — Device Level

Represents user context & intent · multimodal local decisions · lives inside Octahedron (O8) nodes

IDENTITY & SECURITY

Trust Is Structural, Not Procedural

Manufacturing-Time Identity

Every device gets a network-native identity at manufacture — region, tier, role, cryptographic root of trust.

Topology-Constrained Authority

Devices don't select destinations. Connection emerges only when identity is structurally admissible.

Zero Intermediate Intervention

Each E2E link occupies a physically isolated optical core. No routing, no switching — zero interception surface.

Connection Lifecycle

1. Identity Provisioned
2. Topology Binding
3. Presence Detection
4. Edge Validation
5. Path Formation
6. Post-Formation Attestation

MARKET OPPORTUNITY

A Multi-Trillion Dollar Inflection Point

$4.5T
Global Telecom Infrastructure (2030E)

$2.1T
AI Infrastructure Market (2030E)

$98B
6G Network Market (2035E)

$23B
Embodied AI Market (2030E)

Why Now?

→ 6G standardization is creating a once-in-a-generation architectural window — decisions now define the next 30 years.

→ Edge AI & embodied AI demand sub-millisecond, deterministic connectivity — packet switching cannot deliver this.

→ Digital-sovereignty concerns are accelerating demand for non-centralized, verifiable infrastructure.

STANDARDIZATION

Advancing Through ITU-T Standards

FG-AINN — Contributions I-236-R1 & I-237-R1: structural embedded AI, projection-based sovereign governance, optical-native architecture (8th meeting, 2026).

FG-EAI — Contribution I-011-R1: network-layer interface for embodied AI — per-device 8-channel isolated E2E, deterministic parallel delivery.

FG-AI4SSC — Smart-city contribution: Cube Router as candidate Minimum Interoperability Mechanism (MIM); accountable cross-system interoperability.

SG13 Referral — Two independent new-Question tracks requested for ITU-T SG13; tri-lateral liaison framework across SG13 · SG20 · SG21 proposed.

Status: contributions submitted and in progress through ITU-T Focus Group procedures.

COMPETITIVE ADVANTAGE

Why Velsanet Cannot Be Replicated

CAPABILITY	INTERNET / 5G	PROPRIETARY SDN	VELSANET
Packet-free E2E	✗	Partial	✔ Physical
Topology-first Design	✗	✗	✔ Native
AI Embedded in Network	✗	✗	✔ 3 Layers
Manufacturing Identity	✗	Partial	✔ Always
Sovereign Governance	✗	✗	✔ Q2–Q4
ITU Standards Engagement	Incumbent	✗	✔ Active

TECHNOLOGY ROADMAP

From White Paper to Global Infrastructure

Phase 1
2026

- MOCT prototype hardware
- PAI + AAI integration lab
- City-scale pilot design
- ITU standards engagement

Phase 2
2027

- First city E2E center
- 3-layer AI deployment
- Identity provisioning

Phase 3
2028

- Multi-city mesh
- National sovereign Q4
- Partner onboarding

Phase 4
2029+

- Continental Q3 federation
- IVGF governance
- Global VELSA layer

23 white papers completed · Core IP established · ITU standards track active · Ready for partnership & investment

DEPLOYMENT FOUNDATION

A Real-World Proving Ground

Velsanet is designed to be proven in a dedicated research-and-validation hub — an all-in-one environment where international researchers, enterprises and institutions co-develop, test and demonstrate next-generation network technology.

- **Research & living infrastructure** for resident international researchers
- **E2E testbed** for first real-world implementation
- **Open participation** — governments, enterprises, academia, IGOs

Government Engagement

Discussions are underway with Korean government bodies regarding a national special zone as a candidate site for an international joint-research and demonstration hub.

Early-stage; under multi-agency review.

INVESTMENT THESIS

Why Velsanet, Why Now

1 • First-Principles Architecture — Velsanet doesn't patch the Internet; it replaces the foundation. 23 interlocking white papers define a complete system.

2 • AI-Native from Day One — intelligence is a structural property of the network, not a retrofit.

3 • Timing: The 6G Window — architectural decisions in 2026–2028 will define connectivity for 30 years.

4 • Standards-Track Credibility — active contributions across three ITU-T Focus Groups, with SG13 referral requested.

5 • Defensible Moat — topology-first design creates structural barriers packet-switching competitors inherit permanently.

LET'S BUILD THE FUTURE TOGETHER

VELSANET

The Network that Thinks, Learns & Grows.

We are seeking strategic partners and investors to bring Velsanet's vision from white paper to global deployment.

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WHITE PAPERS	AI LAYERS	STANDARDS TRACK	SCALABILITY

Contact us to receive the full technical white paper package.